

## Data Flow Diagram

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|----------------------|--------------------------|
| <b>Date</b>          | 29 September 2026        |
| <b>Team ID</b>       | 6ab4ce760fc666a751b52130 |
| <b>Project Name</b>  | AI StudyBuddy            |
| <b>Maximum Marks</b> | 2 Marks                  |

### Data Flow Diagram

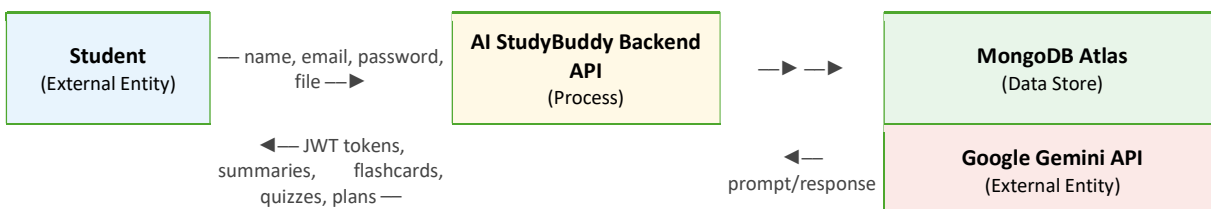
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

### DFD Symbol Legend

| Symbol                            | Name            | Description                                                                                                                 |
|-----------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
| Oval / Rounded shape              | External Entity | A person, organization, or system outside the project that sends or receives data (e.g., Student, Admin, Google Gemini API) |
| Rectangle with numbered header    | Process         | An activity that transforms incoming data into outgoing data (e.g., Authenticate User, Generate AI Summary)                 |
| Rectangle (solid fill, no number) | Data Store      | A place where data is held for later use (e.g., MongoDB Users Collection, MongoDB Materials Collection)                     |
| Labeled arrow                     | Data Flow       | The movement of data between entities, processes, and data stores                                                           |

### Level 0 DFD — AI StudyBuddy (Context Diagram)

Shows the overall system as a single process with external entities and high-level data flows.



### Level 1 DFD — Detailed Data Flows

Breaks down the system into four main processes showing detailed data movement between entities, processes, and data stores.

| External Entity              |   | Process                 |   | Data Store / External                         | Data Flows                                                                                                                                                  |
|------------------------------|---|-------------------------|---|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Student<br>(External Entity) | ⇌ | P1: Authenticate User   | ⇌ | D1: Users Collection (MongoDB)                | IN: name, email, password, role<br>STORE: hashed password, user data<br>OUT: JWT access token, refresh token                                                |
| Student<br>(External Entity) | ⇌ | P2: Manage Materials    | ⇌ | D2: Materials Collection (MongoDB)            | IN: file (.txt/.md/.pdf), title<br>STORE: title, content, filename, userId<br>OUT: material list / details                                                  |
| Student<br>(External Entity) | ⇌ | P3: Generate AI Content | ⇌ | Google Gemini API (External Entity)           | IN: materialId, count, goal, hoursPerDay, days<br>PROMPT: material content to Gemini<br>STORE: AI-generated content to D2<br>OUT: AI-generated content JSON |
| Admin<br>(External Entity)   | ⇌ | P4: Admin Operations    | ⇌ | D1: Users Collection D2: Materials Collection | IN: list users / delete user / get stats<br>QUERY: users, materials count<br>OUT: user list / stats / deletion confirmation                                 |

Data Flow Details

| Flow | From              | To                       | Data                                             |
|------|-------------------|--------------------------|--------------------------------------------------|
| 1    | Student           | P1: Authenticate User    | name, email, password, role                      |
| 2    | P1                | D1: Users Collection     | hashed password, user data                       |
| 3    | P1                | Student                  | JWT access token, refresh token                  |
| 4    | Student           | P2: Manage Materials     | file (.txt/.md/.pdf), title                      |
| 5    | P2                | D2: Materials Collection | title, content, filename, userId                 |
| 6    | Student           | P3: Generate AI Content  | materialId, count, goal, hoursPerDay, days       |
| 7    | P3                | Google Gemini API        | prompt with material content                     |
| 8    | Google Gemini API | P3                       | AI-generated text (summary/flashcards/quiz/plan) |
| 9    | P3                | D2: Materials Collection | save generated content to material record        |
| 10   | P3                | Student                  | AI-generated content JSON response               |
| 11   | Admin             | P4: Admin Operations     | list users / delete user / get stats request     |
| 12   | P4                | D1 + D2                  | query users, materials count                     |
| 13   | P4                | Admin                    | user list / stats / deletion confirmation        |